Visionary

11+3 · pack included

Compose RL + RFT for 2026 BEHAVIOR
No Discord / Drive

Kanta not seated — all five packs built

Spine: unlock PPO on flow VLAs; BEHAVIOR still prefers RFT

Chen / Liu / Zhang et al. · RLinf · ICML-style (TENTATIVE)

Club spine · Oct 5

Unlock PPO on flow VLAs · BEHAVIOR still prefers RFT

BLOCKED

Flow VLAs

intractable $\log\pi$
+ deterministic ODE
→ no PG / no explore

→

π_RL OPENS

Flow-Noise / SDE

tractable Gaussian
+ PPO/GAE on ~300M
expert (VLM frozen)

→

FAST SIM

LIBERO / ManiSkill

+27–31 pp ID avg
few-shot+RL > full SFT
partial env OOD

→

SLOW SIM

BEHAVIOR / OmniG

Comet cites π_RL
then uses RFT
sample-efficiency gate

When to use π_RL

- Parallel sparse-success sim available
- OpenPI π_0 / $\pi_{0.5}$ (or GR00T) already SFT'd
- Want ID specialists beyond demos
- Can afford 8×H100-class + 64–320 envs
- Real deploy = sim RL → ODE eval (not real RL)

When to stay on RFT

- Slow household sims (BEHAVIOR)
- Heterogeneous GPU / RT-core split pain
- Need coverage flywheel, not on-policy
- Comet path: reject→SFT (Q 0.224→0.345)
- 2026: compose RFT then short π_RL burst

Online RL (π_RL) vs RFT (Comet) · product fork

Axis	Online RL (π_RL)	RFT (Comet)
Needs	logπ + exploration	success filter only
Sample efficiency	low	higher for slow sims
Compute topology	tight env↔train loop	embar. parallel eval OK
Ceiling	can exceed demos	capped by reachable modes
Cross-task OOD	weak (ML45)	also limited
BEHAVIOR'25	cited, not used	Q 0.224 → 0.345
2026 upside	if faster eval arrives	safe default flywheel

2026 compose bets · Visionary

No Discord / Drive — arXiv + RLinf docs + Comet only

1 **RFT then short π_RL**

Coverage cheaply → residual failure modes once rewards / faster eval exist

2 **Don't freeze VLM forever**

LoRA-on-VLM + action RL for household visual diversity

3 **Tune H' for advantages**

Replan horizon = credit quality, not only SFT smoothness

4 **Smarter hybrid τ**

TempFlow / Mix-GRPO timing ideas (TENTATIVE App I)

5 **Dual $\pi_{0.5}$ + GR00T**

Same adapter on both 2026 organizer baselines

6 **Eval-first politics**

Reward design = equal research to algorithm novelty

Lineage arc · Archaeologist

flow VLAs blocked PG → π_RL opens it → slow-sim routes around via RFT

2023	2024	2025	blocked	π_RL	fork
DP / ACT	**π_0**	**$\pi_{0.5}$ OpenPI**	**no logπ**	**Noise/SDE**	**RFT**
chunk+replan	flow VLA	open weights	ODE = no explore	PPO unlocked	Comet path

Priors absorbed (CLEAR)

π_0 / $\pi_{0.5}$ · Lipman FM / Rectified Flow · ReinFlow · Flow-GRPO · DPPO

PPO / GAE · RLinf / RLinf-VLA · Song PF-ODE↔SDE · 3DGS (Real2Sim)

Contemporaries: SimpleVLA-RL, RL4VLA, FPO (not compared) · Subsequent: Comet RFT, BEHAVIOR'25 forks

Do not fabricate: no BEHAVIOR win · no ICML acceptance · no real online RL at scale

Product choice · Stakeholder

Canonical open online-RL adapter for flow VLAs — when you own fast sim

WHAT YOU GET

- OpenPI $\pi_0/\pi_{0.5}$ via RLinf
- Few-shot+RL can beat full SFT %
- +27–31 pp ID on 4 suites
- GR00T path in appendix
- Open code + HF weights

WHAT IT COSTS

- H100-class parallel sim
- 64–320 envs typical
- Reward engineering
- Not true real online RL
- Sample inefficiency admitted

ASK OF LEADERSHIP

- π_RL if fast sim + sparse success rewards exist
- RFT if BEHAVIOR-slow
- Budget for twin sims before chasing online PPO on OmniG

Decision rule (say aloud)

Prefer π_RL with fast parallel sim; prefer RFT when BEHAVIOR-slow.

Comet already made this call for 2025 — cite then skip online RL.

Real2Sim2Real 40% = prototype, not a product claim.

OOD honesty · what transfers

TRANSFERS

- ManiSkill visual / layout / exec shifts
- CALVIN ABC→D scene transfer
- Partial relative OOD gains (~+100%)
- Preserves SFT OOD skills (less forgetting)

DOES NOT

- MetaWorld ML45 novel tasks — oscillates
- Cross-task skill invention via RL
- Open-world discovery narrative
- Visual OOD ceiling (frozen VLM)

Reading for Reviewer + Empiricist

RL = action-level refinement on seen skills

Not a substitute for multi-task demos or stage structure

BEHAVIOR implication: recovery / temporal efficiency — not 50-task invention

Control note · not classical planner

Flow → MDP

Stochastic via Flow-SDE
or Flow-Noise train path
Deploy = ODE Euler

Hybrid sampler

1 SDE step / env step
rest ODE → ~2× faster
vs full two-layer

PPO on expert

~300M action expert
VLM frozen default
chunk-level reward

Chunk length H = credit-assignment knob

Paper: π_0 LIBERO H=50 · $\pi_{0.5}$ H=10 · ManiSkill H=8 · CALVIN/MW H=5

Larger chunks ↑ SFT baseline but ↓ explained variance / RL ceiling

Empiricist: prefer Comet-like mid horizons + receding H' over cargo-cult H=50

Tattoo numbers · Empiricist must memorize

π_0 SFT → Noise

51.1 → 80.3

+29.2 pp ID avg

$\pi_{0.5}$ SFT → SDE

55.6 → 86.6

+31.0 pp ID avg

LIBERO Long

43.9 → 94.0

$\pi_{0.5}$ Flow-Noise

Few-shot+RL

98.3% > 96.9%

beats full $\pi_{0.5}$ SFT

Real2Sim2Real

SFT fail → 40%

3DGS · thin n

Extras

PPO>GRPO ~6pp

GR00T 52.5→89.9

Also tattoo

ManiSkill $\pi_{0.5}$ 40.1→90.9 IND · OOD ~26→49–53 · CALVIN Len-5 61.3→87.0 · ML45 novel tasks: NO stable gain

Honesty: 'beats full SFT' omits interactive sample cost · OOD = env/visual, not open-world discovery

GPU honesty · Empiricist / Stakeholder

Workload	VRAM	Wall	Smoke
Unit-test Flow-SDE	<2 GB	minutes	Y
Eval-only K×a grid	12–24 GB	hours	Y
Critic fit	8–20 GB	hours	Y
Tiny PPO ≤8 envs	24–48 GB	1–3 days	Partial
Paper LIBERO-style	8×H100	days–weeks	N
BEHAVIOR online RL	multi extreme	out of scope	N

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https://github.com/RLinf/RLinf https://huggingface.co/RLinf

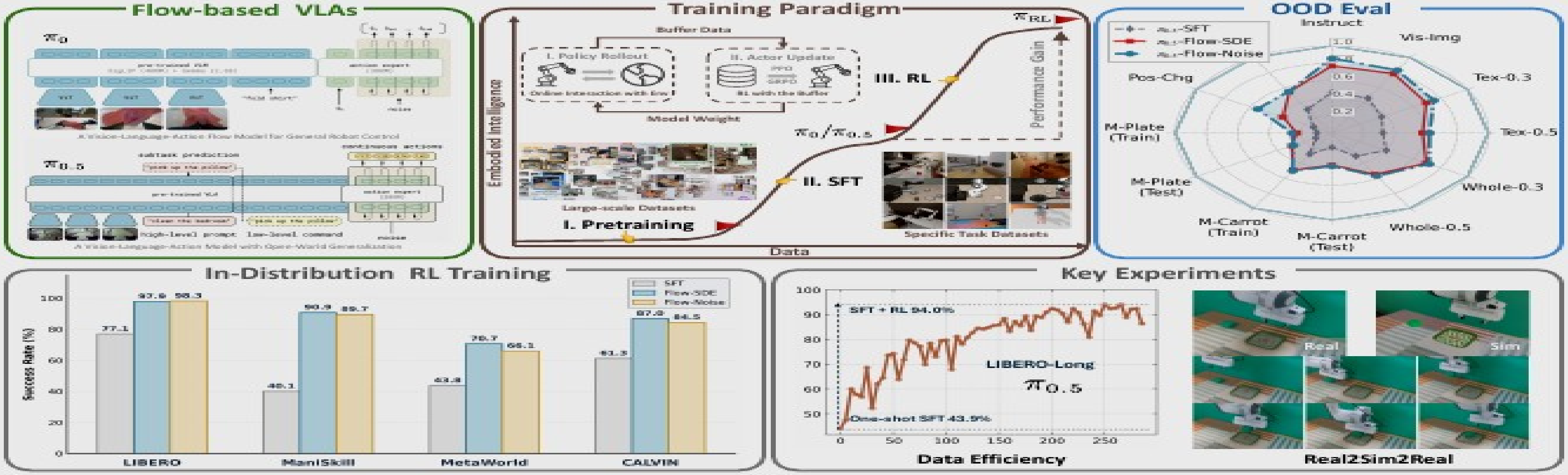

Figure 1. π_{RL}: An online RL framework for flow-based VLAs. Incorporating two solutions, Flow-Noise and Flow-SDE, π_{RL} enhances the performance and generalization of SFT-aligned models across extensive ID benchmarks and OOD settings. Refined with RL, few-shot SFT policies achieve performance comparable to full dataset baselines. Additionally, we facilitate seamless zero-shot sim-to-real transfer by constructing a simulator with 3D Gaussian Splatting as the rendering engine to narrow the visual domain gap.

Abstract

Vision-Language-Action (VLA) models enable robots to understand and perform complex tasks from multimodal input. Although recent work explores using reinforcement learning (RL) to automate the laborious data collection process in scaling supervised fine-tuning (SFT), applying RL to large-scale flow-based VLAs (*e.g.*, π_0, $\pi_{0.5}$) remains challenging due to intractable action log-likelihoods raised from flow matching. We address this challenge with π_{RL}, featuring two technical approaches: (1) **Flow-Noise** models the denoising process as a discrete-time MDP with a learnable noise network for exact log-likelihood computation. (2) **Flow-SDE** integrates denoising with agent-environment interaction, formulating a two-layer MDP that employs ODE-to-SDE conversion for efficient RL exploration. We evaluate π_{RL} across various benchmarks, with experiments demonstrating that RL yields significant performance improvements in both in-distribution and out-of-distribution settings.

[1]Tsinghua University [2]Peking University [3]Institute of Automation, Chinese Academy of Sciences [4]Carnegie Mellon University [5]Infinigence AI [6]Zhongguancun Academy ♯ Work done at Tsinghua University ∗ Equal Contributions . Correspondence to: Yu Wang <yu-wang@tsinghua.edu.cn>, Chao Yu <zoeyuchao@gmail.com>.

Preprint. January 30, 2026.

Ruthless smoke priority · defer full online RL

Empiricist matrix — P0 Y / P1 Partial / Defer N

P	Smoke	Experiment	What
P0	Y	**Released ckpt eval**	OpenPI / RLinf weights on 1 suite vs SFT
P0	Y	**Eval-only K×a grid**	a∈{0.2,0.5,0.8} · K∈{1,2,4,8} · no PPO
P0	Y	**Unit-test Flow-SDE**	Toy dim · ODE vs SDE marginal · hybrid
P1	Partial	**Offline AW-CFM**	Advantage-weighted CFM on logged rollouts
P1	Partial	**Critic warm-up**	V_vlm vs V_expert on fixed buffer
P1	Partial	**Tiny PPO ≤8 envs**	≤20–40 updates · not Tab.1 reproduction
Defer	N	**Tab.1 multi-bench**	8×H100 · 4 suites · hundreds of epochs
Defer	N	**ManiSkill 4k / Real / BEHAVIOR**	Weeks + hardware — out of quiet scope

Visionary close

π_RL = open online half; RFT = pragmatic offline half — compose them. Skip Discord/Drive.

π_RL · arXiv 2510.25889 · Oct 5 · Q&A

Unlock PPO on flow VLAs · BEHAVIOR prefers RFT

π_{RL}: Online RL Fine-tuning for Flow-based Vision-Language-Action Models

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